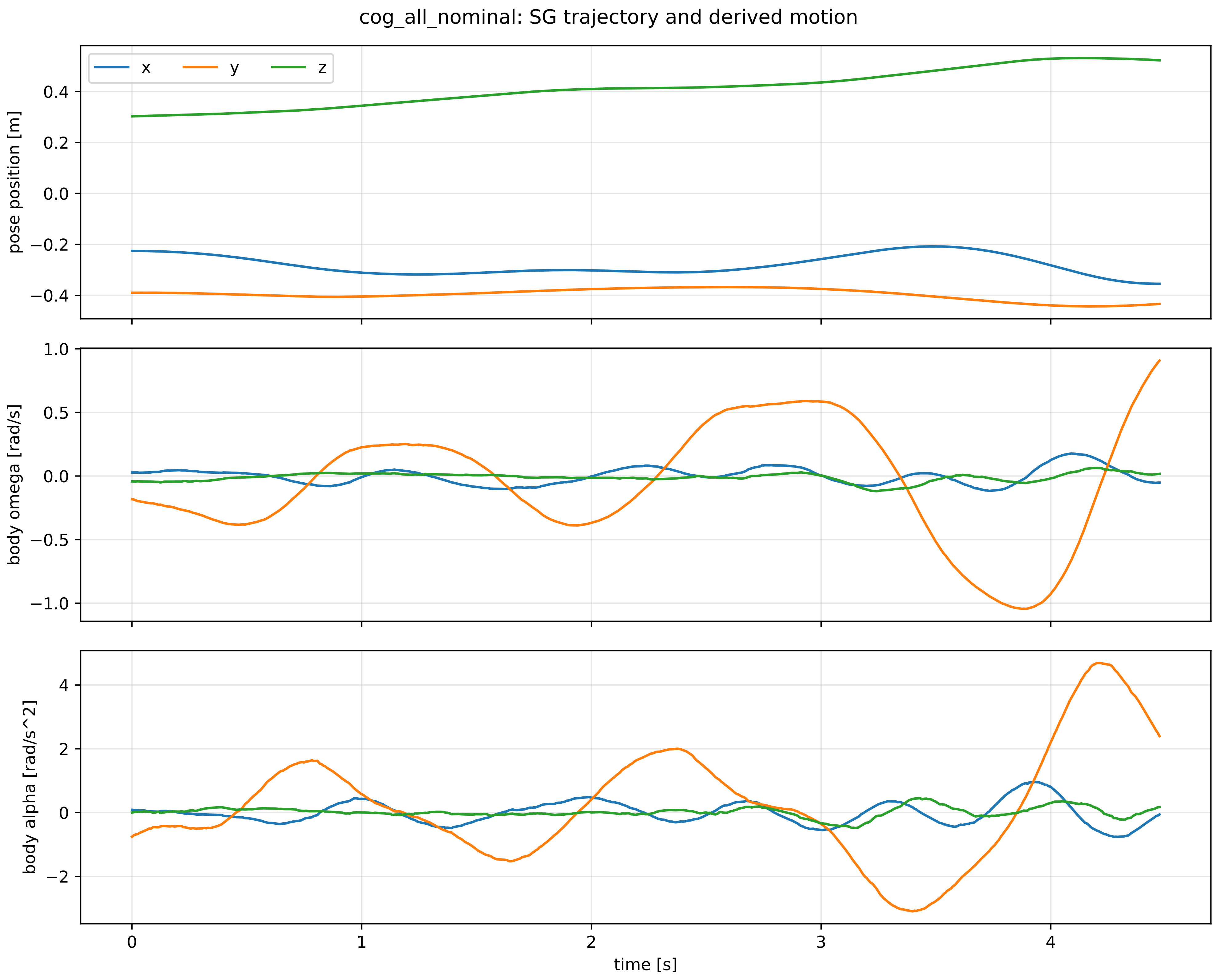
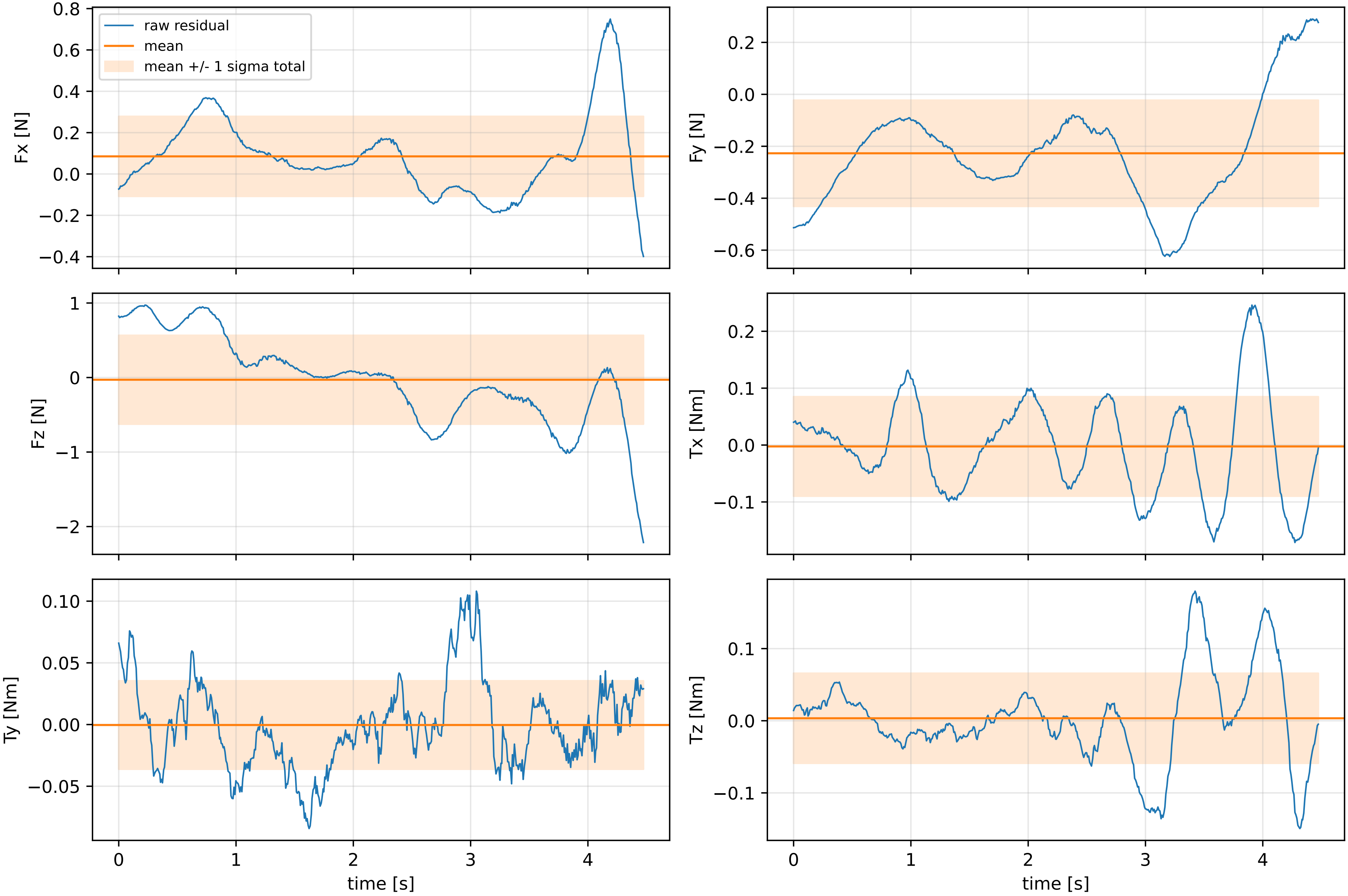
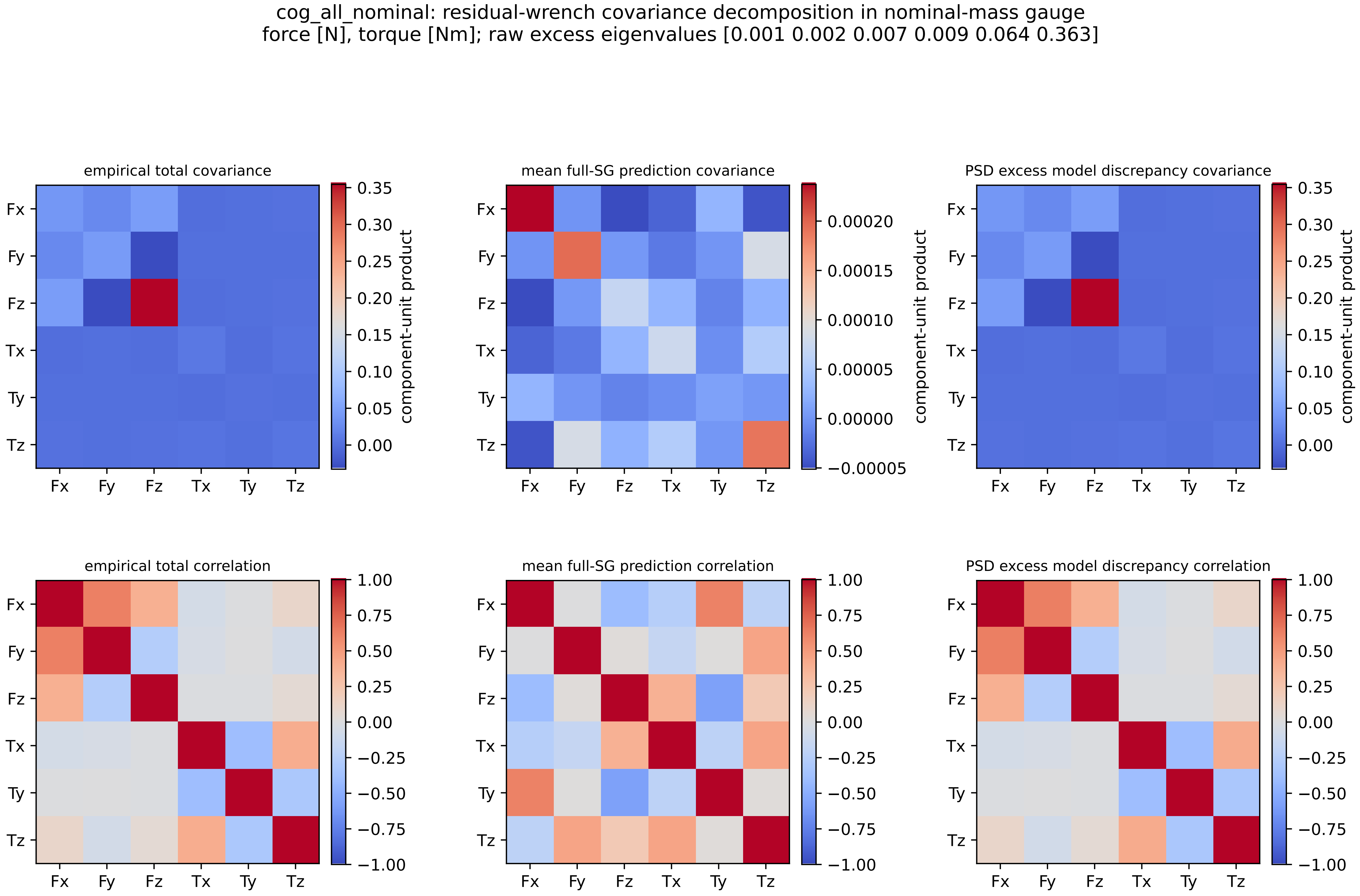
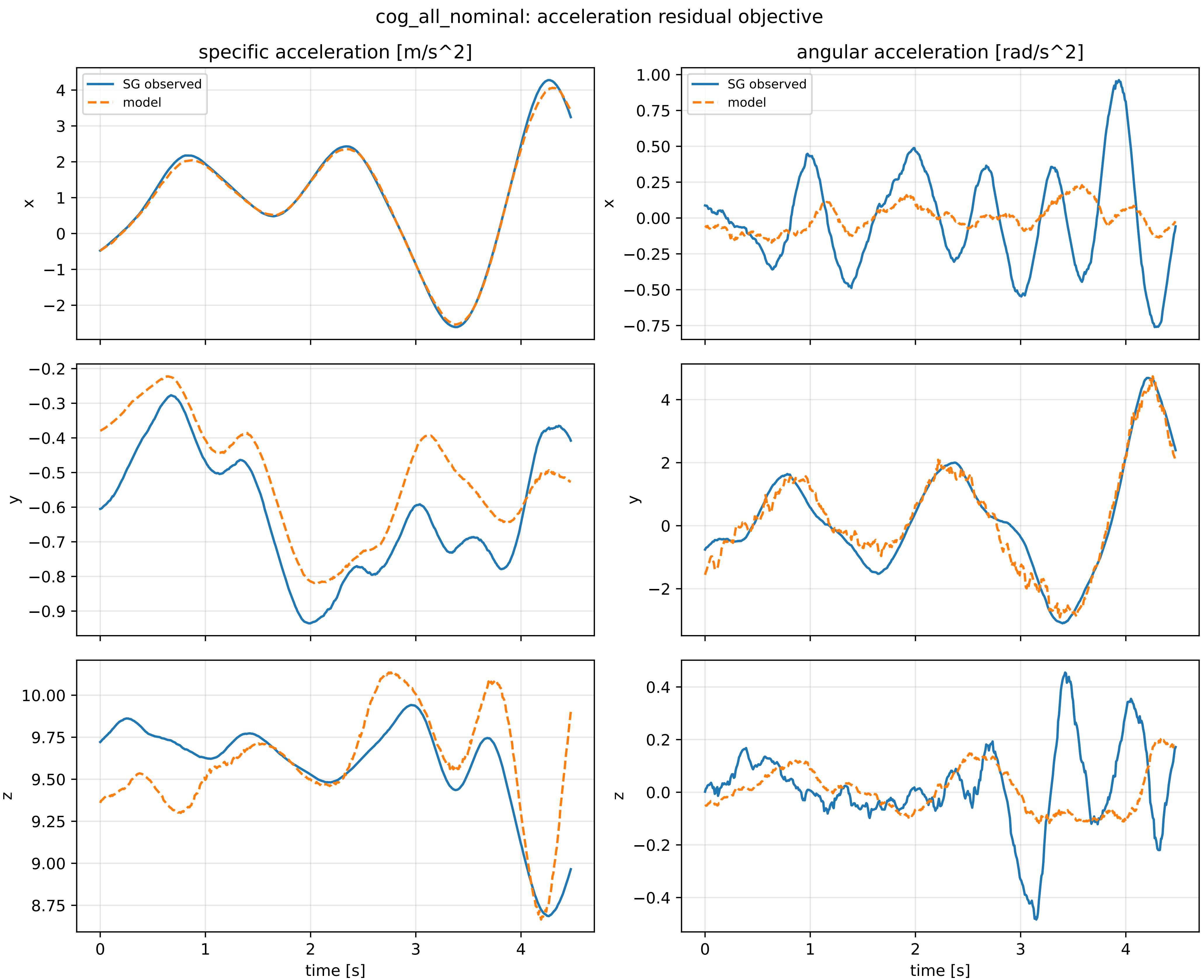

cog_all_nominal: SG trajectory and derived motion
x
y
z
pose position [m]
0.4
0.2
0.0
−0.2
−0.4
body omega [rad/s]
1.0
0.5
0.0
−0.5
−1.0
body alpha [rad/s^2]
4
2
0
−2
0
1
2
3
4
time [s]

cog_all_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg

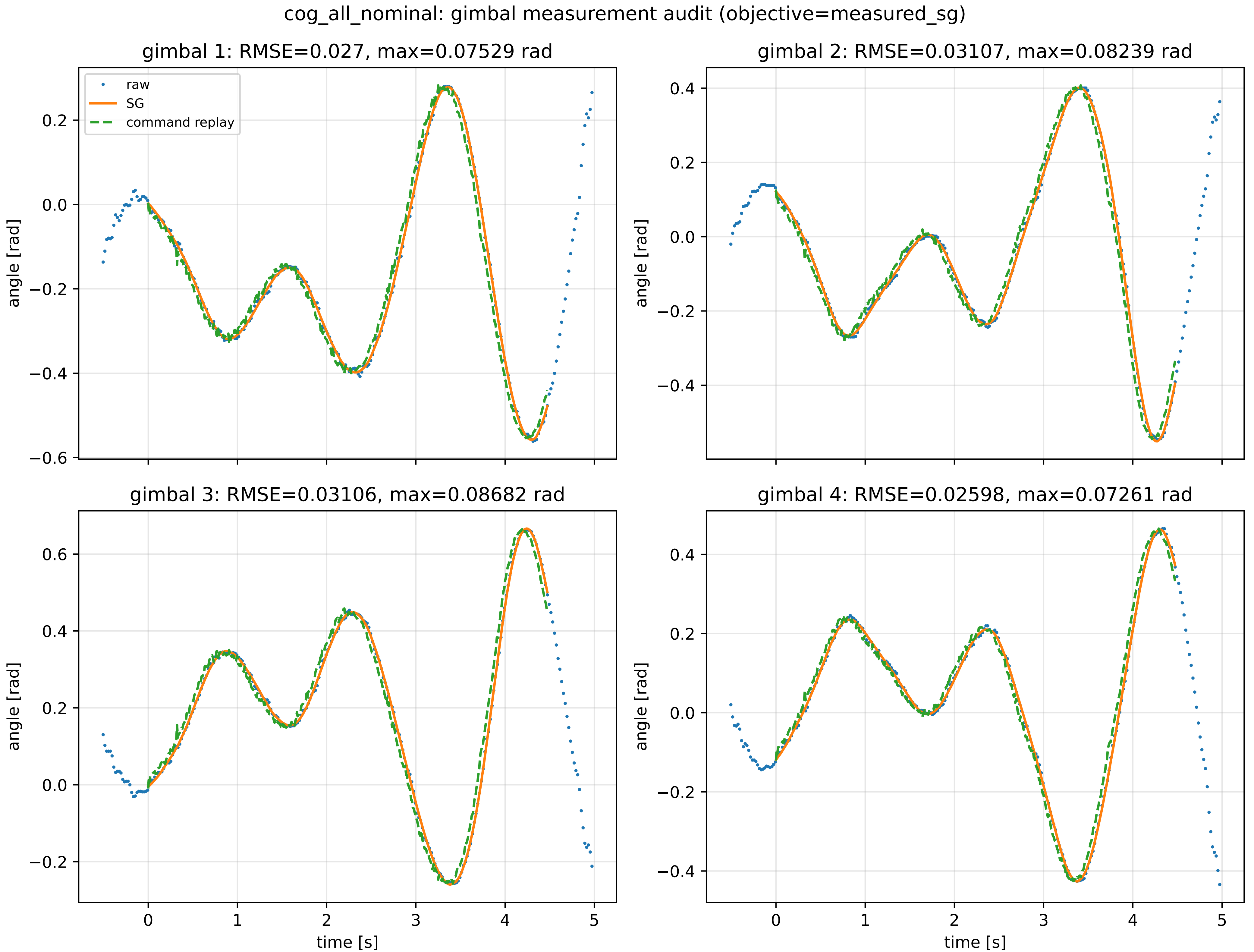


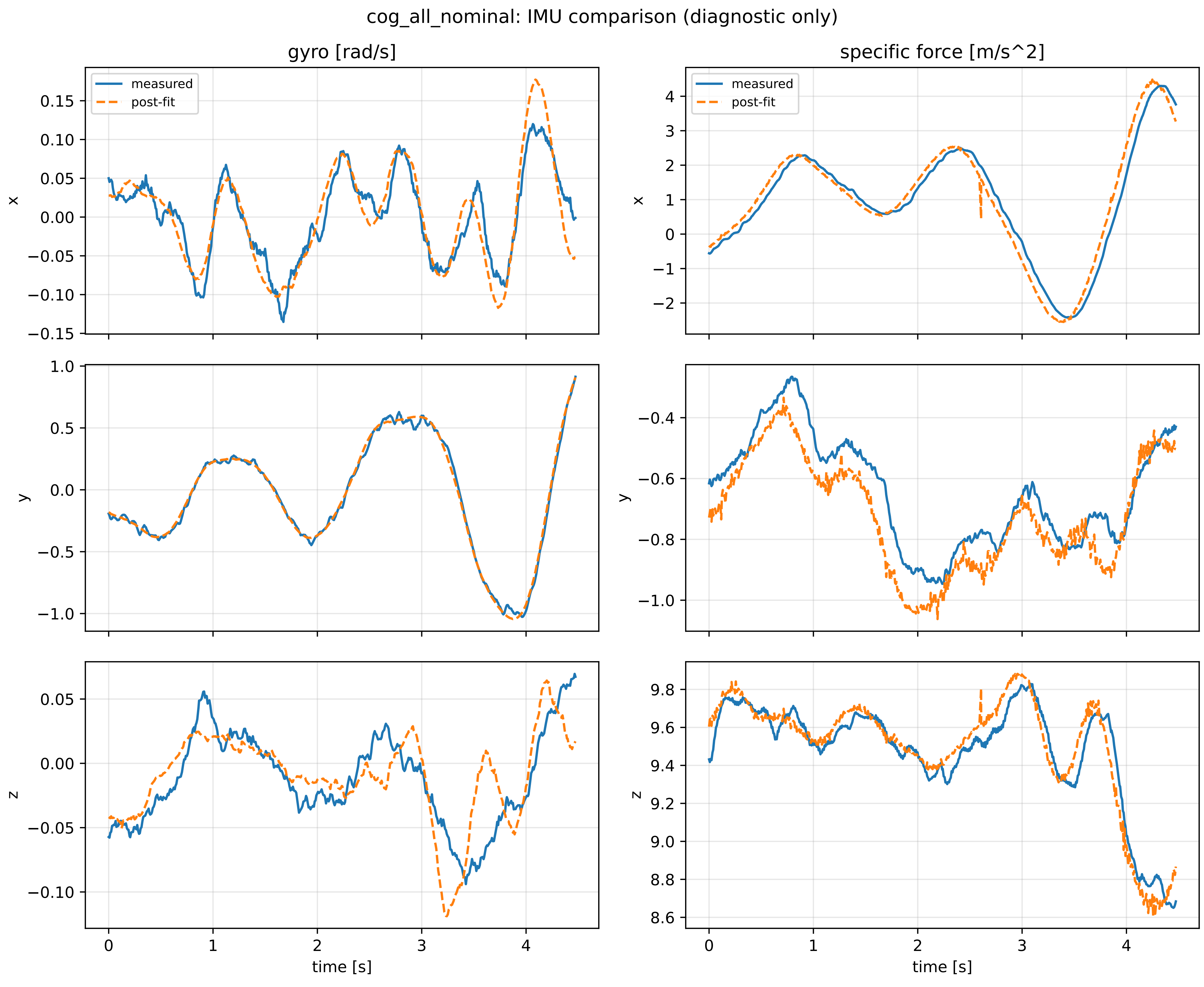
vector RMS about zero: force 0.70172 N, torque 0.11309 Nm; component std about mean: force [0.1932 0.2042 0.5954], torque [0.0873 0.0358 0.0623]

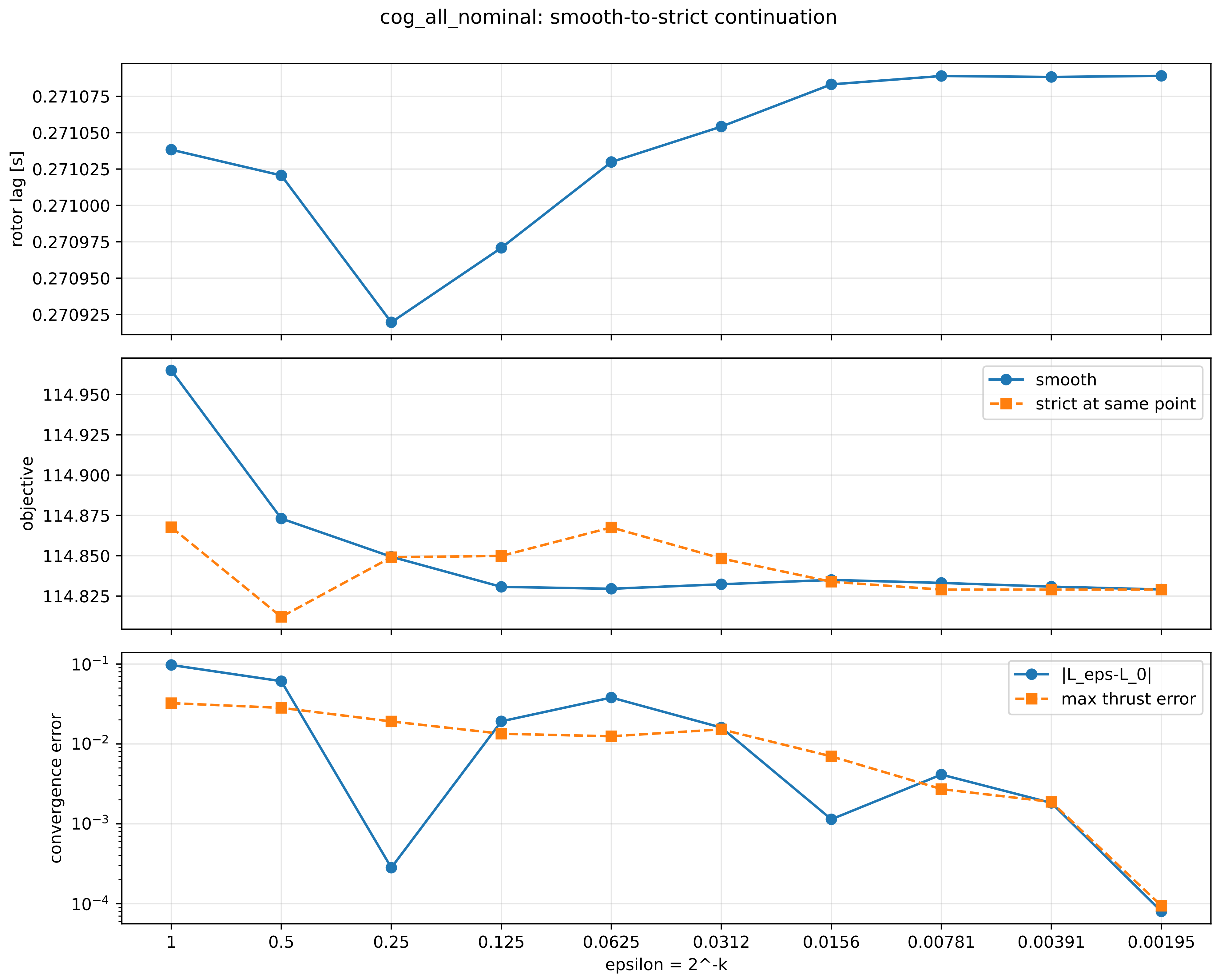

cog_all_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.001 0.002 0.007 0.009 0.064 0.363]
empirical total covariance
mean full-SG prediction covariance
PSD excess model discrepancy covariance
empirical total correlation
mean full-SG prediction correlation
PSD excess model discrepancy correlation
Fx
Fy
Fz
Tx
Ty
Tz
component-unit product
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0.00020
0.00015
0.00010
0.00005
0.00000
−0.00005
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

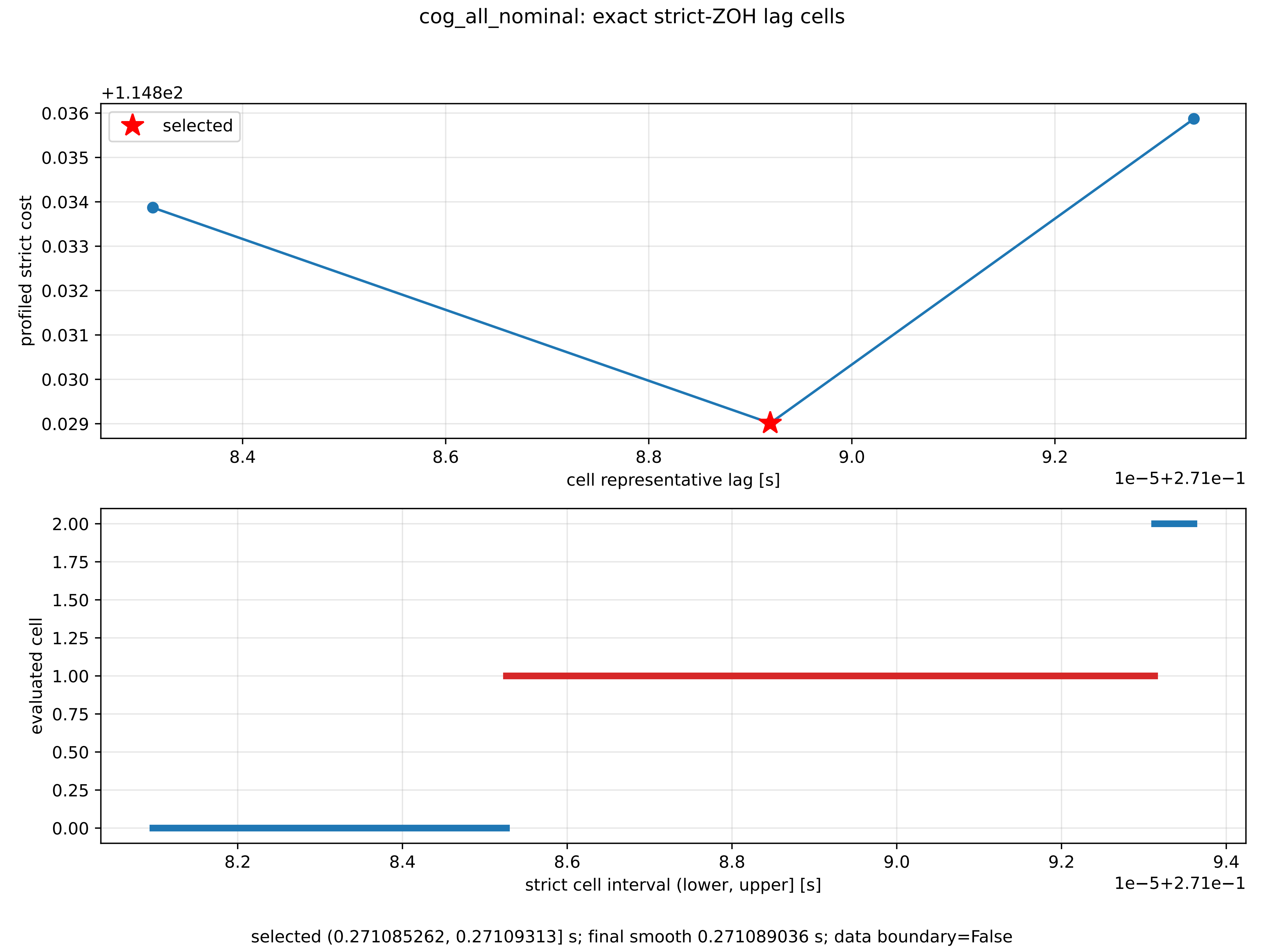


cog_all_nominal: acceleration residual objective
specific acceleration [m/s^2]
angular acceleration [rad/s^2]
SG observed
model
x
y
z
time [s]


cog_all_nominal: gimbal measurement audit (objective=measured_sg)
gimbal 1: RMSE=0.027, max=0.07529 rad
gimbal 2: RMSE=0.03107, max=0.08239 rad
gimbal 3: RMSE=0.03106, max=0.08682 rad
gimbal 4: RMSE=0.02598, max=0.07261 rad
raw
SG
command replay
angle [rad]
time [s]

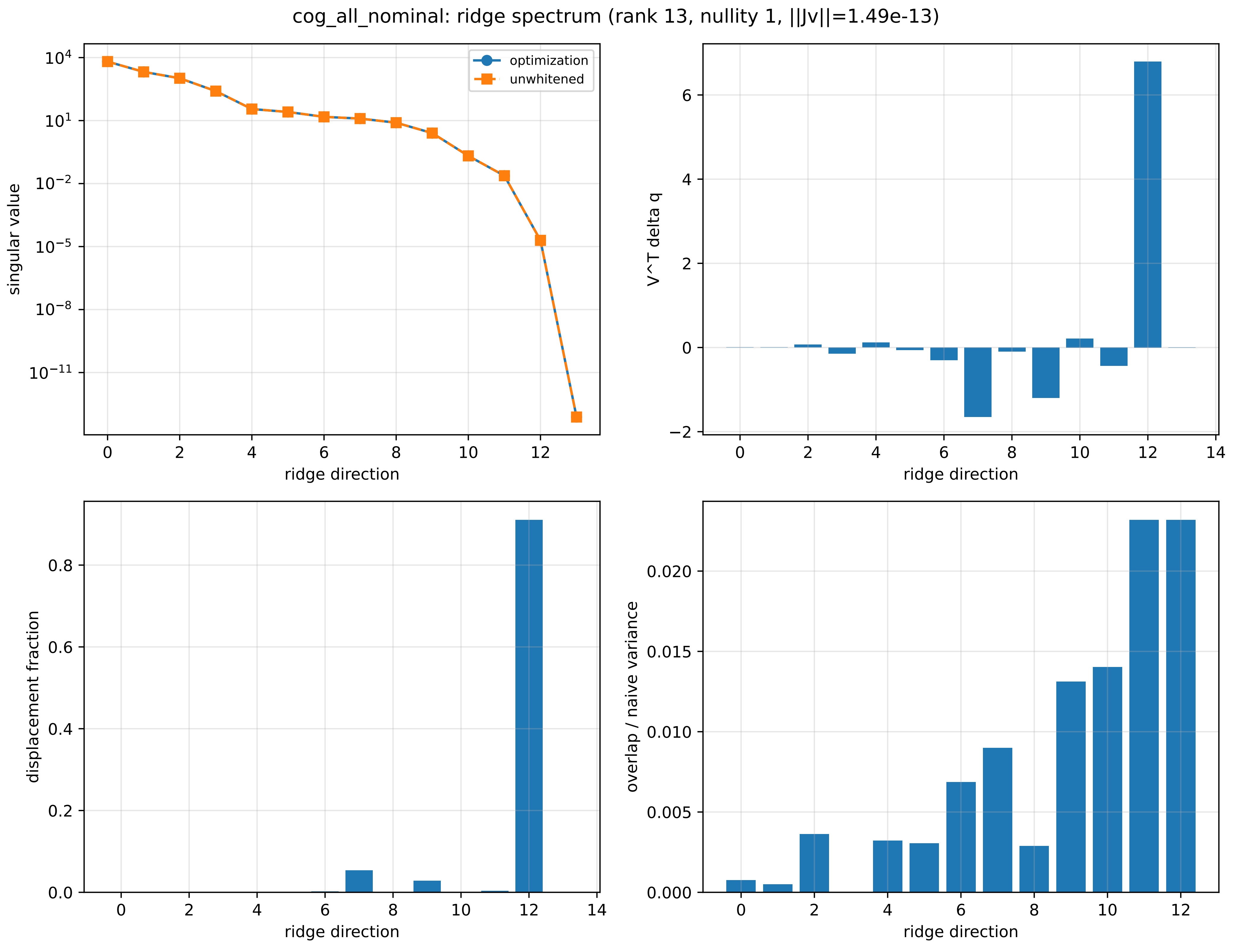

cog_all_nominal: IMU comparison (diagnostic only)
gyro [rad/s]
specific force [m/s^2]
measured
post-fit
x
y
z
time [s]


cog_all_nominal: smooth-to-strict continuation
rotor lag [s]
0.271075
0.271050
0.271025
0.271000
0.270975
0.270950
0.270925
objective
114.950
114.925
114.900
114.875
114.850
114.825
smooth
strict at same point
convergence error
10^{-1}
10^{-2}
10^{-3}
10^{-4}
|L_eps-L_0|
max thrust error
1
0.5
0.25
0.125
0.0625
0.0312
0.0156
0.00781
0.00391
0.00195
epsilon = 2^-k


cog_all_nominal: exact strict-ZOH lag cells
+1.148e2
selected
profiled strict cost
0.036
0.035
0.034
0.033
0.032
0.031
0.030
0.029
8.4
8.6
8.8
9.0
9.2
cell representative lag [s]
1e−5+2.71e−1
evaluated cell
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
8.2
8.4
8.6
8.8
9.0
9.2
9.4
strict cell interval (lower, upper] [s]
1e−5+2.71e−1
selected (0.271085262, 0.27109313] s; final smooth 0.271089036 s; data boundary=False

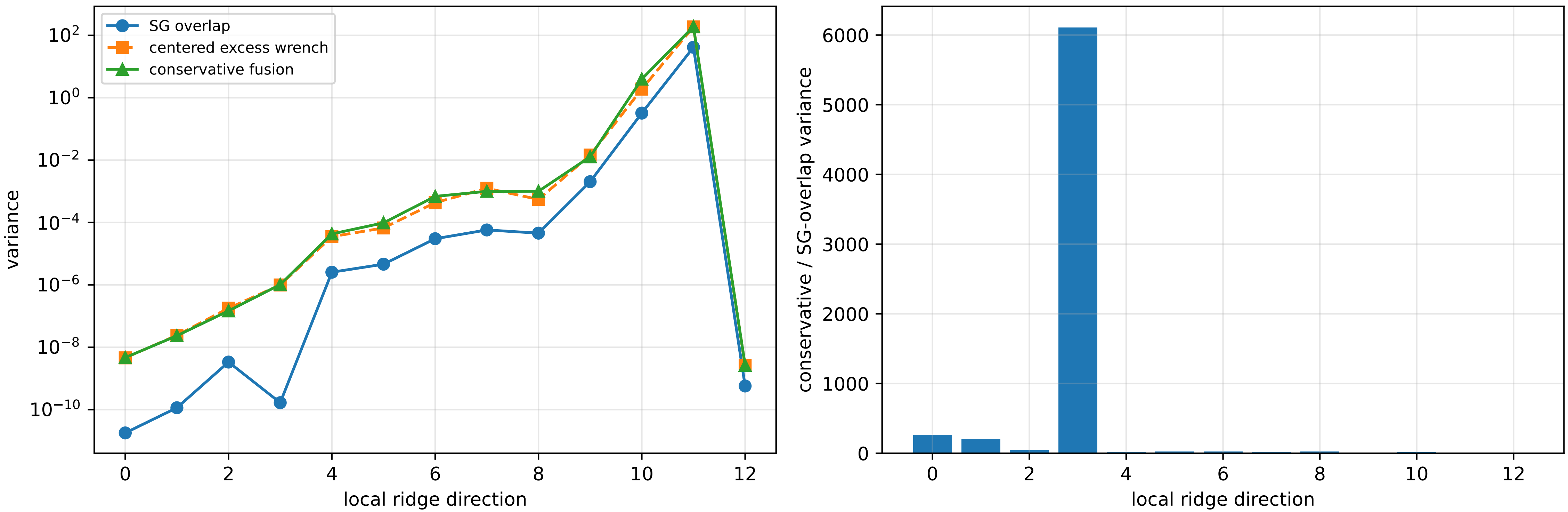
cog_all_nominal: parameter estimate

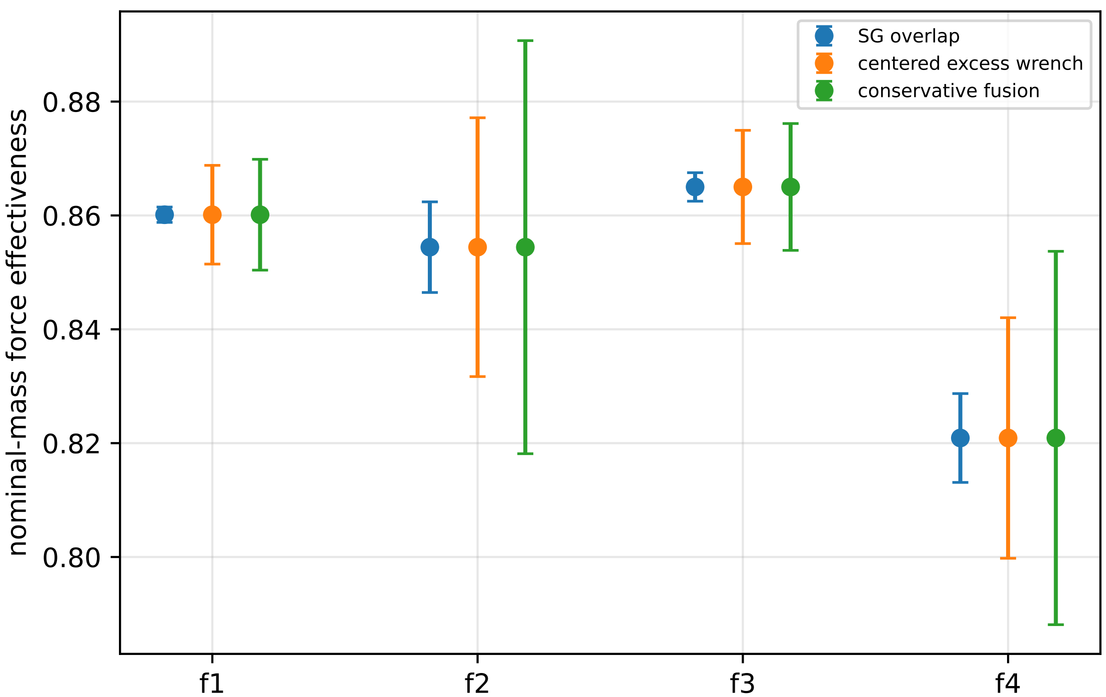
```
Case: cog_all_nominal
status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 4.05191491
inertia [kg m^2]:
[[ 0.44463106  0.00599693  0.02360596]
 [ 0.00599693  0.1417382  -0.01304697]
 [ 0.02360596 -0.01304697  0.57759031]]
CoG body [m]: [-2.02472065e-03 -3.05231839e-05  9.50976561e-03]
force effectiveness: [1.48203985 1.47221684 1.49044461 1.41445807]

scale-free J/m [m^2]:
[[ 0.10973356  0.00148002  0.00582588]
 [ 0.00148002  0.03498055 -0.00321995]
 [ 0.00582588 -0.00321995  0.14254749]]
scale-free f/m: [0.36576283 0.36333854 0.36783709 0.34908385]

rotor lag cell: (0.271085262, 0.27109313] s
representative: 0.271089196 s
```



cog_all_nominal: ridge spectrum (rank 13, nullity 1, ||Jv||=1.49e-13)
optimization
unwhitened
singular value
ridge direction
V^T delta q
displacement fraction
overlap / naive variance

cog_all_nominal: Conservative fusion uncertainty

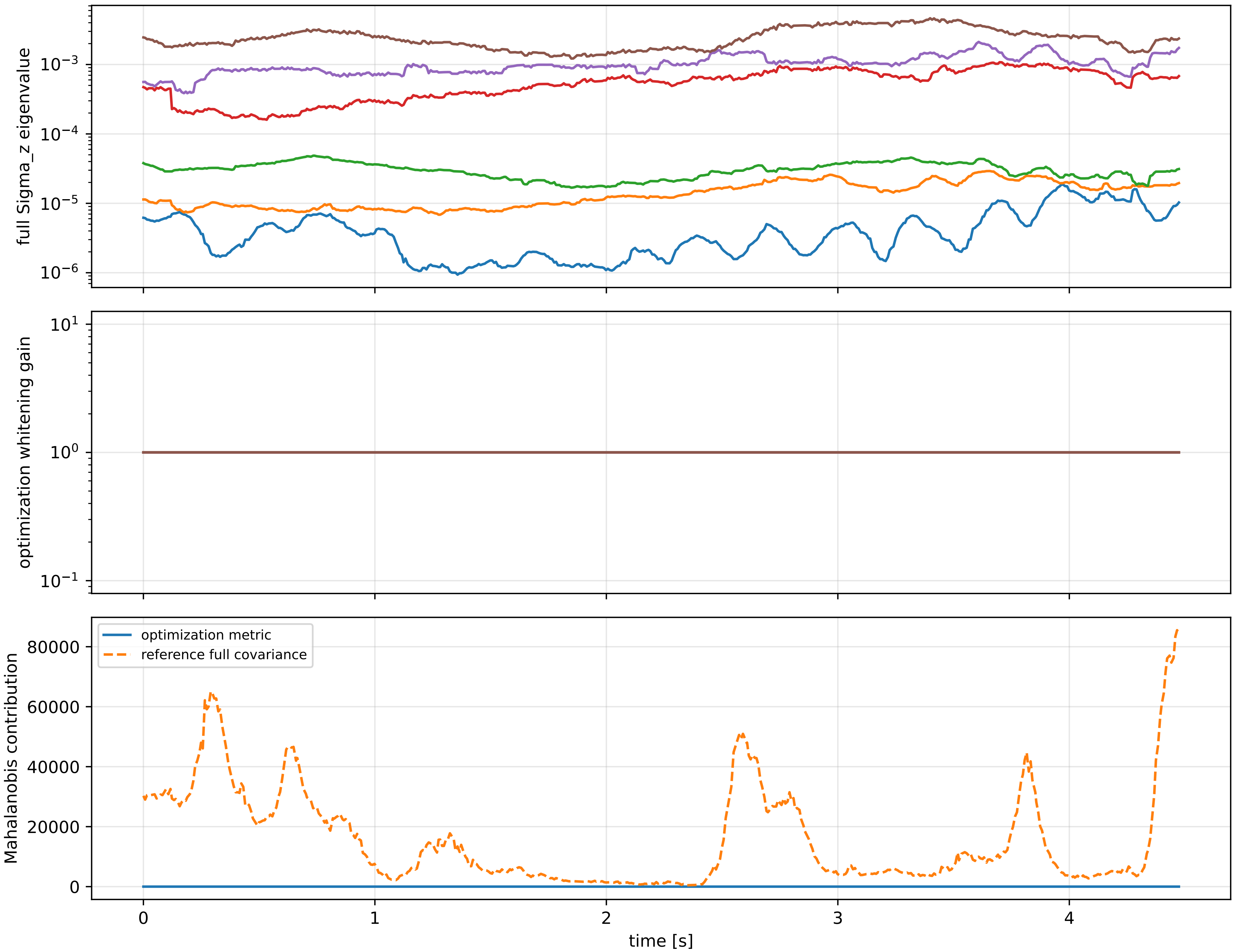


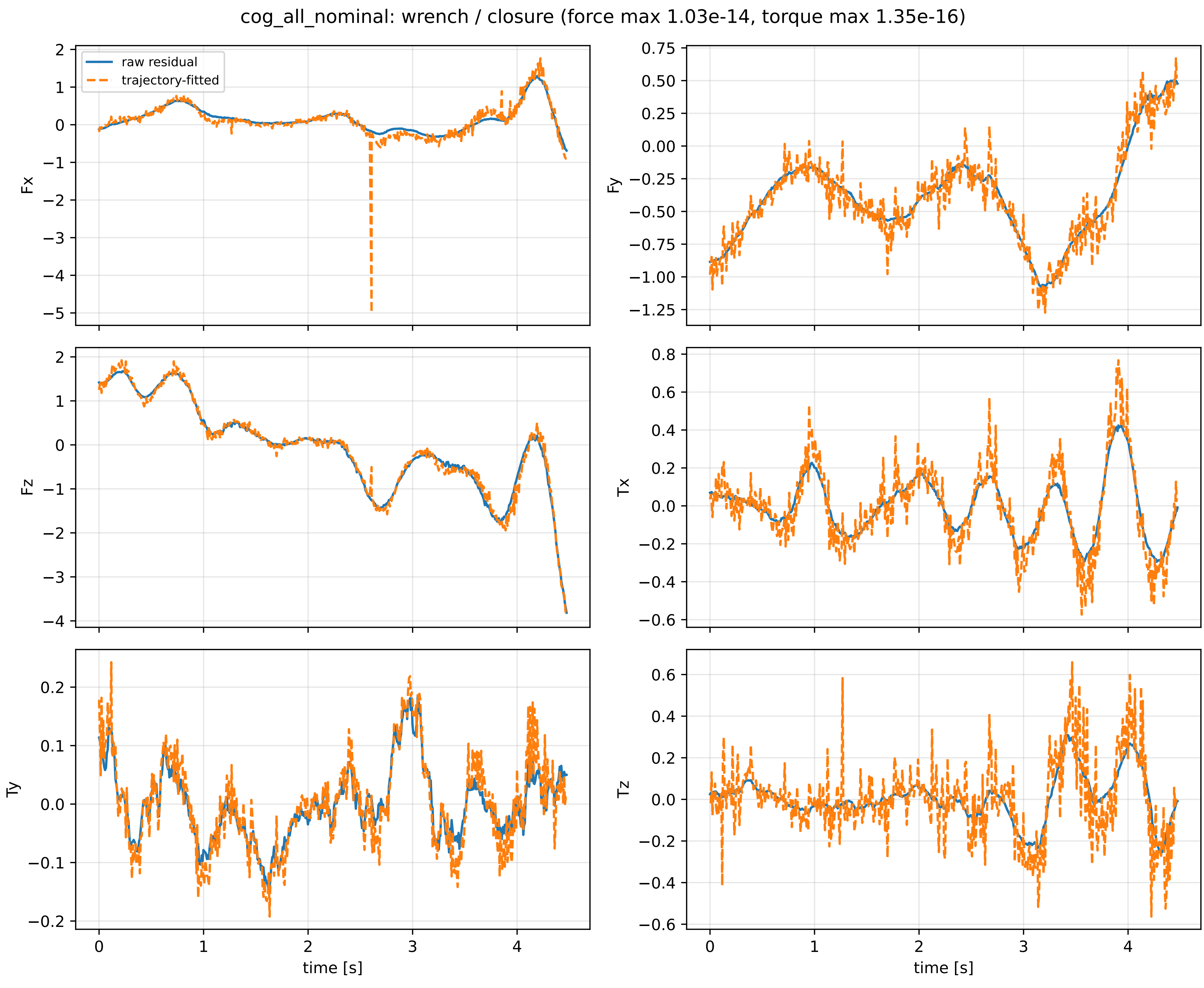



```
Top three non-gauge ambiguous directions
tr(M_res) / tr(M_SG) = 233.8
wrench-to-acceleration recovery max error = 2.55e-15
exact scale gauge ridge index = 13 (alignment 1)

index 11: variance=190.7, ratio=4.592
  log_mass=+0.0407; second_moment_diag_1=+0.516
  second_moment_diag_2=-0.017; second_moment_diag_3=-0.703
  second_moment_offdiag_12=+0.148; second_moment_offdiag_13=-0.374
  second_moment_offdiag_23=+0.262; cog_x=+2.18e-05
  cog_y=-1.43e-05; cog_z=-2.96e-07
  log_force_eff_1=+0.0407; log_force_eff_2=+0.0409
  log_force_eff_3=+0.0407; log_force_eff_4=+0.0405

index 10: variance=3.972, ratio=12.35
  log_mass=-0.0139; second_moment_diag_1=+0.0171
  second_moment_diag_2=-0.207; second_moment_diag_3=+0.259
  second_moment_offdiag_12=+0.743; second_moment_offdiag_13=+0.212
  second_moment_offdiag_23=+0.54; cog_x=-0.00142
  cog_y=+0.0018; cog_z=-7.35e-06
  log_force_eff_1=-0.0147; log_force_eff_2=-0.0292
  log_force_eff_3=-0.0117; log_force_eff_4=-0.000283

index 9: variance=0.01283, ratio=6.296
  log_mass=+0.158; second_moment_diag_1=+0.0461
  second_moment_diag_2=-0.892; second_moment_diag_3=+0.056
  second_moment_offdiag_12=-0.0443; second_moment_offdiag_13=-0.106
  second_moment_offdiag_23=-0.247; cog_x=-0.00265
  cog_y=+0.0011; cog_z=+0.000246
  log_force_eff_1=+0.164; log_force_eff_2=+0.142
  log_force_eff_3=+0.15; log_force_eff_4=+0.176
```

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

cog_all_nominal: optimization vs reference covariance





cog_all_nominal: wrench / closure (force max 1.03e-14, torque max 1.35e-16)
raw residual
trajectory-fitted
Fx
Fy
Fz
Tx
Ty
Tz
time [s]
time [s]

cog_all_nominal: optional parameter-prior audit

```
Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.
name: pseudo_condition_cog_all_nominal
role: pseudo_conditioning_ablation
source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/group/cog_all_nominal.
SHA256: c8e09152b70075c9dea19c9858059b27978f15fa2ccc061d3e475ff0b4b7566a

data objective:  114.829010674
prior objective: 2.06975936245e-06
total objective: 114.829012744

This configuration uses an intentionally tight artificial standard deviation
for an ablation-style pseudo-conditioning experiment; it is not a calibrated
physical uncertainty.

factor: cog_all_nominal (cog_position_body_m)
  components: x, y, z
  target: [-2.02470856e-03 -3.05265789e-05  9.50974960e-03]
  std:    [1.e-05 1.e-05 1.e-05]
  final:  [-2.02472065e-03 -3.05231839e-05  9.50976561e-03]
  error:  [-1.20911429e-08  3.39506275e-09  1.60071761e-08]
  standardized residual: [-0.00120911  0.00033951  0.00160072]
  factor objective: 2.06975936245e-06
```